SCIENCE 1.4 **Name:** ***ANSWERS***

WORKSHEET FOUR **WRITING AND NAMING FORMULAE**

TABLE OF IONS

1^+	2^+	3^+	1^-	2^-
H^+	Mg^{2+}	Al^{3+}	Cl^-	O^{2-}
Na^+	Ca^{2+}	Fe^{3+}	OH^-	CO_3^{2-}
K^+	Fe^{2+}		NO_3^-	SO_4^{2-}
	Cu^{2+}		HCO_3^-	
	Zn^{2+}			
	Pb^{2+}			

1. Complete the following chart by writing the formula formed by pairing up the positive ion down the side with the negative ion across the top. The first one has been done for you.

	O^{2-}	Cl^-	NO_3^-	SO_4^{2-}
Na^+	***Na_2O***	***$NaCl$***	***$NaNO_3$***	***Na_2SO_4***
	Sodium oxide	***Sodium chloride***	***Sodium nitrate***	***Sodium sulfate***
Zn^{2+}	***ZnO***	***$ZnCl_2$***	***$ZnNO_3$***	***$ZnSO_4$***
	Zinc oxide	***Zinc chloride***	***Zinc nitrate***	***Zinc sulfate***
Fe^{3+}	***Fe_2O_3***	***$FeCl_3$***	***$Fe(NO_3)_3$***	***$Fe_2(SO_4)_3$***
	Iron(III) oxide	***Iron(III) chloride***	***Iron(III) nitrate***	***Iron(III) sulfate***
Mg^{2+}	***MgO***	***$MgCl_2$***	***$Mg(NO_3)_2$***	***$MgSO_4$***
	Magnesium oxide	***Magnesium chloride***	***Magnesium nitrate***	***Magnesium sulfate***
K^+	***K_2O***	***KCl***	***KNO_3***	***K_2SO_4***
	potassium oxide	***potassium chloride***	***potassium nitrate***	***potassium sulfate***

2. Complete the following chart by writing the formula formed by pairing up the positive ion down the side with the negative ion across the top. The shaded compounds are not stable.

	Cl^-	OH^-	HCO_3^-	CO_3^{2-}
H^+	**HCl**	***H_2O***	***H_2CO_3***	***H_2CO_3***
	hydrogen chloride	***Water***	***Carbonic acid***	***Carbonic acid***
Ca^{2+}	***$CaCl_2$***	***$Ca(OH)_2$***	***$Ca(HCO_3)_2$***	***$CaCO_3$***
	Calcium chloride	***Calcium hydroxide***	***Calcium bicarbonate***	***Calcium carbonate***
Pb^{2+}	***$PbCl_2$***	***$Pb(OH)_2$***		***$PbCO_3$***
	Lead chloride	***Lead hydroxide***		***Lead carbonate***
Al^{3+}	***$AlCl_3$***	***$Al(OH)_3$***		***$Al_2(CO_3)_3$***
	Aluminium chloride	***Aluminium hydroxide***		***Aluminium carbonate***
Fe^{2+}	***$FeCl_2$***	***$Fe(OH)_2$***		***$FeCO_3$***
	Iron(II) chloride	***Iron(II) hydroxide***		***Iron(II) carbonate***

3. Write the formulae for the following compounds.

(a) iron(III)hydroxide ***$Fe(OH)_3$*** (b) magnesium carbonate ***$MgCO_3$***

(c) potassium bicarbonate ***$KHCO_3$*** (d) zinc hydroxide ***$Zn(OH)_2$***

(e) hydrogen sulphate ***H_2SO_4*** (f) lead nitrate ***$Pb(NO_3)_2$***

(g) aluminium oxide ***Al_2O_3*** (h) iron(II)sulphate ***$FeSO_4$***

(i) hydrogen nitrate ***HNO_3*** (j) zinc carbonate ***$ZnCO_3$***

(k) carbon dioxide ***CO_2*** (l) hydrochloric acid ***HCl***

(m) sodium bicarbonate ***$NaHCO_3$*** (n) hydrogen oxide ***H_2O***

(o) sodium hydroxide ***NaOH*** (p) calcium sulphate ***$CaSO_4$***